Quantitative Macro Theory **HW #2**
Spring 2019 **Due at 3pm on Sat 2/2/2019**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (100 points) Consider this problem:

$$V(k_i, \varepsilon_r) = \max_{c,k',l,h} \frac{1}{1-\gamma}\left(c - \frac{l^{1+\theta}}{1+\theta}\right)^{1-\gamma} + \beta \sum_{s=1}^{2} \pi_{rs} V(k'_j, \varepsilon_s),$$

subject to,

$$c = A(k_i h)^{\alpha} l^{1-\alpha} - k'_j e^{-\varepsilon_r} + k_i \left(1 - h^{\omega}/\omega\right) e^{-\varepsilon_r},$$

Calibration

Set $\beta = 0.96$ as in Kydland and Prescott (1982), $\alpha = 0.29$ to match the capital's share of income (NIPA), $1/\theta = 1.7$ to match the intertemporal elasticity of labor supply estimated by MaCurdy (1981) and others, $\gamma = 2$, and $\omega = 1.42$ to imply a rate of depreciation of 10% in equilibrium.

In terms of the stochastic process, set

$$\pi_{11} = \pi_{22} = \pi \text{ and } \varepsilon_1 = -\varepsilon_2 = \Theta.$$

Note that the standard deviation $\sigma = \Theta$ and the coefficient of autocorrelation is $\lambda = 2\pi - 1$. These parameters are picked so as to make the model generate the same standard deviation and first-order serial correlation for output as is observed in the data. $\sigma = 0.051$ and $\lambda = 0.44$.

(Note: I think I covered all the parameters but if you think I am missing one, pick the value of your choice and let me know.)

(a) (20 points) Solve the model using value function iteration + discretization. Plot V and k' as a function of k and include two lines, one for each value of the shock ε. Implement the tricks that we talked about in class that take advantage of the fact that the policy function is increasing and the maximand is concave. How much time did you save with each of them? Implement the Howard iteration approach to save time. Report how much time did you save and what was the optimal value of n_h that you found.

(b) (20 points) Look for data for time series of output, consumption, investment, hours worked, productivity, and capital utilization for the US, Canada, and Germany for the postwar period. Do a table like Table 1 and report the sources very carefully. Before computing the statistics detrend the series using the HP method or other. What are the key differences and similarities across countries?

Table 1: Key data moments

Variable	Standard deviation	Correlation with output	Autocorrelation
Output			
Consumption			
Investment			
Hours worked			
Productivity			
Utilization			

(c) (20 points) Compute the stationary distribution of capital and ε iterating starting with a uniform distribution. Plot the initial distribution, the distribution after 5 iterations, and the steady state distribution for both capital and ε. Use this distribution to compute the statistics in Table 1 and compare your finding with data. Can the model account for the main findings? What are the greatest failures? Explain.iterating starting with a uniform distribution. Plot the initial distribution, the distribution after 5 iterations, and the steady state distribution for both capital and ε. Use this distribution to compute the statistics in Table 1 and compare your finding with data. Can the model account for the main findings? What are the greatest failures? Explain.

(d) (20 points) Compute the stationary distribution of capital and ε by simlating a time series of 10,100 period and disregarding the first 100 periods. Plot the distribution for both capital and ε and compare them with the findings in the previous question.

(e) (20 points) Compute a path for all the series in Table 1 starting from the average value of capital and the high realization of ε. For the first 4 periods ε stays high and then drop to the low value and stay there for 8 periods. A total of 12 quarters. Do plots Comparing the evolution of the series in the model with the data you obtained in the last question of HW 1 but now starting in the third quarter of 2006 until the second quarter of 2009 (12 quarters). What is the greatest success of the model in term of matching the data? How do the results compare with the ones in HW 1, in which the shock was a TFP? Explain the main differences in the behavior across models with different shocks.